

Messaging Application Pseudocode

1. App Start

```
START APP

CONNECT to database
INITIALIZE server
ENABLE routes and WebSocket server
WAIT for requests
```

2. Register User

```
FUNCTION register(username, password):

    IF username already exists:
        RETURN error "User exists"

    HASH password

    CREATE new user in database

    RETURN success
```

3. Login User

```
FUNCTION login(username, password):

    FIND user in database

    IF user not found:
        RETURN error

    COMPARE password with hashed password

    IF incorrect:
        RETURN error

    GENERATE auth token

    RETURN token
```

4. Create Conversation

```
FUNCTION createConversation(userId, otherUserId):

    CHECK both users exist

    CREATE new conversation
```

```
ADD userId to Participant table
ADD otherUserId to Participant table

RETURN conversationId
```

5. Send Message

```
FUNCTION sendMessage(senderId, conversationId, content):

    CHECK sender is part of conversation

    CREATE message in database:
        - content
        - senderId
        - conversationId
        - timestamp

    EMIT message via WebSocket to all participants

    RETURN success
```

6. Receive Message (Real-Time)

```
ON WebSocket connection:

    STORE userId with socket

    WHEN message is sent:

        FIND all participants in conversation

        FOR EACH participant:
            IF participant is online:
                SEND message through socket
```

7. Fetch Conversations

```
FUNCTION getUserConversations(userId):

    FIND all conversations where user is a participant

    RETURN list of conversations
```

8. Fetch Messages

```
FUNCTION getMessages(conversationId):

    FIND all messages for conversation
    SORT by timestamp ASC
```

```
RETURN messages
```

9. Notifications (Offline Users)

```
WHEN message is sent:
```

```
  FOR EACH participant:
```

```
    IF user is offline:
```

```
      STORE notification in database
```

```
      OR trigger push notification
```

10. Logout

```
FUNCTION logout(userId):
```

```
  REMOVE user socket connection
```

```
  INVALIDATE token (optional)
```

```
  RETURN success
```